Kimia Mohammadnezhad

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Education

M.Sc in Geomatics – Remote Sensing Engineering 2019–2022
K. N. Toosi University of Technology, Tehran, Iran GPA: 3.77/4

Thesis: Integration of epidemiology, remote sensing, and GIS to analyze spatial patterns of stomach cancer in Golestan Province.
Fused multi-temporal satellite imagery (Hyperion, Landsat) and meteorological data for complex spatial-pattern analysis using advanced geospatial and ML techniques.

B.Sc in Civil – Geomatics Engineering 2015–2019
K. N. Toosi University of Technology, Tehran, Iran GPA: 3.03/4

Thesis: Assessment of drought dynamics across Iran
using two decades of MODIS satellite observations, focusing on spatial and temporal drought patterns.

Research Experience

Remote Research Collaboration — Brown University (Advisor: Dr. Christopher Horvat) 2023–Present

- Developing transfer-learning approaches for SAR-based sea-ice classification, extracting high-resolution surface features and morphological states essential for understanding local sea-ice dynamics.
- Processed and analyzed multi-sensor polar remote sensing datasets to investigate sea ice variability, concentration uncertainties, and surface conditions.
- Investigated discrepancies between passive microwave retrievals and altimetry-derived observations to identify systematic biases in Antarctic sea ice concentration products.
- Contributed to conference presentations and manuscript preparation on sea ice remote sensing and machine learning applications.

Publications and Presentations

- Consistent Antarctic sea ice concentration biases from passive microwave satellites resolved via ICESat-2 altimetry - https://doi.org/10.21203/rs.3.rs-6272536/v1
- Rectifying and Understanding Significant Biases in Passive-Microwave-Derived Antarctic Sea Ice Concentration - AGU 2025 Annual Meeting
- Towards Automated SAR Classification of Antarctic Sea Ice via Transfer Learning-Based Machine Learning - AGU 2025 Annual Meeting
- Global gridded products of sea ice concentration, wave climate, and sea ice floe size distribution with ICESat-2: examining marginal ice zone variability and passive microwave uncertainties. AGU 2024
- Modeling Epidemiology Data with Machine Learning Technique to Detect Risk Factors for Gastric Cancer - Journal of Gastrointestinal Cancer - https://doi.org/10.1007/s12029-023-00952-1
- Investigating heavy metals soil contamination state on the rate of stomach cancer using remote sensing spectral features - Environmental Monitoring and Assessment - https://doi.org/10.1007/s10661-023-11234-5

Research Interests

Cryosphere remote sensing	Sea ice dynamics	Multi-sensor data fusion
Polar climate processes	Machine learning in geosciences	Spatial data assimilation

Skills

Programming & Compute — Python (Advanced), MATLAB (Intermediate), Bash

Geospatial & Data Processing — Xarray, SNAPPY, GDAL, GeoPandas, NetCDF/HDF5 handling, Google Earth Engine (Advanced)

Machine Learning — PyTorch, Scikit-learn, Transfer Learning approaches

Remote Sensing & Geomatics — ENVI (Advanced), QGIS (Advanced), SNAP (Advanced), ArcGIS (Intermediate)

High-Performance Computing — OSCAR HPC environment, GPU-accelerated and parallel processing for large EO and ML/DL workflows

Languages — English (Fluent), Persian (Native), French (Intermediate)

Selected Courses

- Microwave Remote Sensing – K. N. Toosi University of Technology
- Digital Signal Processing - EPFL (Coursera)
- Fuzzy Systems and Neural Networks – K. N. Toosi University of Technology
- Oceanography: A Key to Better Understand Our World (Coursera)
- Data Science – IBM (Coursera)
- Writing in Sciences – Stanford University (Coursera)

Honors

- Ranked among the top 20 candidates nationwide in the Iranian graduate entrance examination for Geomatics Engineering.
- Awarded full academic scholarship for both B.Sc and M.Sc studies.
- Ranked among the top 1500 participants in the Iranian national university entrance examination.

References

Dr. Christopher Horvat
Assistant Professor of Earth, Environmental, and Planetary Sciences — Brown University
Email: christopher_horvat@brown.edu

Dr. Mahmood Reza Sahebi
Associate Professor of Geomatics Engineering — K. N. Toosi University of Technology
Email: sahebi@kntu.ac.ir